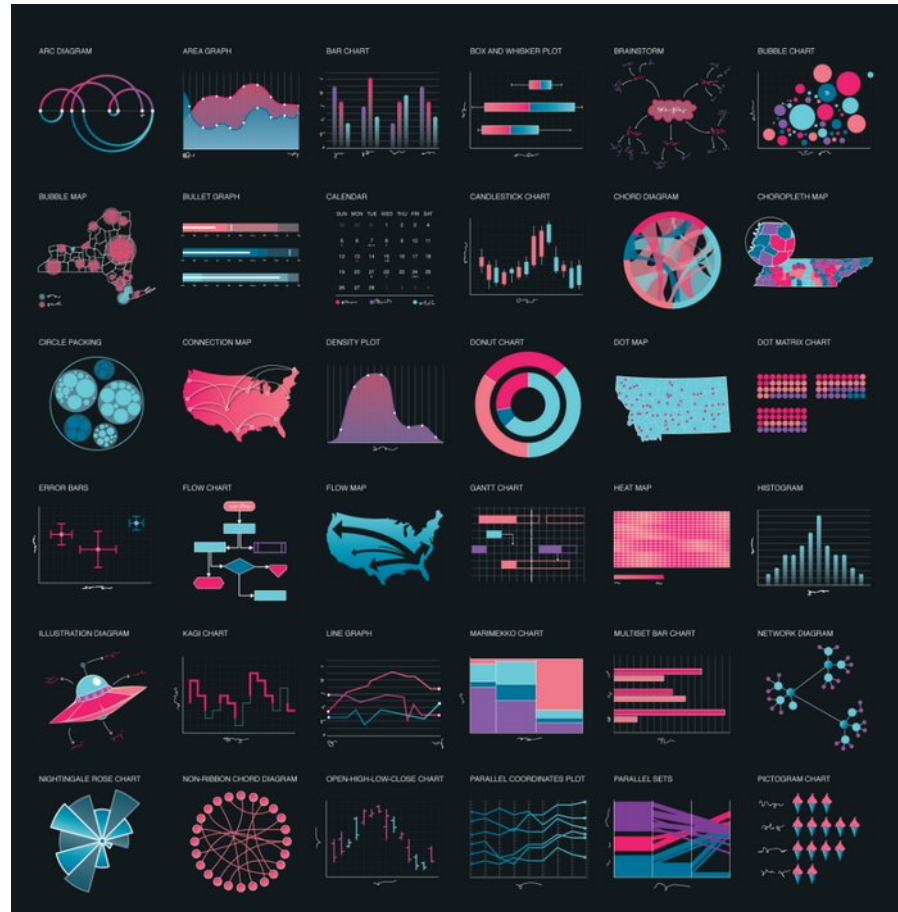




Data Visualization

Prof. Gerry Tsoukalas

Prof. Mohannad Elhamod

Image source: towardsdatascience.com

Boston University Questrom School of Business

When is visualization needed?

Graphing your data and results helps you with:

- Exploration and preliminary analysis:
Finding patterns visually that would not easily stand out otherwise.
- Presentation (reporting & story-telling):
Assess the result of your analysis/modeling and drive the point home.

Why is it a *good idea* to graph your data?

Table 1. Percentage of canceled flights by day of year.

	J	F	M	A	M	J	J	A	S	O	N	D
1	2.1%	2.3%	1.9%	1.5%	1.3%	1.4%	1.6%	1.8%	1.7%	1%	1.4%	2.1%
2	3.2%	2.5%	2.5%	1.3%	1.4%	1.6%	1.3%	1.8%	1.5%	1.3%	1.3%	1.5%
3	3%	2.7%	2.2%	1.6%	1.1%	1.3%	1%	1.7%	1.8%	1.1%	1.1%	1.6%
4	3.2%	3.2%	2.8%	1.5%	1.2%	1.4%	0.9%	1.7%	2.1%	1.4%	1.1%	1.5%
5	2.7%	3%	3.1%	1.1%	1.2%	1.3%	1.2%	1.6%	1.9%	1.6%	1.2%	2.3%
6	3.6%	3.4%	3.5%	1.2%	1.3%	1.6%	1.4%	1.2%	1.9%	1.3%	1.3%	2.1%
7	4.8%	2.1%	2.4%	1.5%	1.2%	1.4%	1.4%	1.4%	1.5%	1.2%	1.1%	1.9%
8	5.2%	2.6%	3%	1.3%	1.2%	1.6%	1.4%	1.3%	1.4%	1.3%	1.1%	2.2%
9	3.5%	2.4%	2.9%	1.6%	1.5%	1.6%	1.6%	1.4%	1.9%	1.2%	0.9%	2.5%
10	2.4%	2.3%	2%	1.7%	1.8%	1.7%	1.8%	1.8%	2%	1.3%	1.3%	2.3%
11	3.4%	3.6%	1.6%	1.9%	1.6%	1.6%	1.5%	1.7%	6.6%	1.3%	1.4%	3.2%
12	3.7%	3.8%	1.6%	1.2%	1.3%	1.9%	1.4%	1.3%	7%	1.1%	1.4%	2.3%
13	3%	3.3%	3.3%	1.1%	1.4%	2.1%	1.5%	1.6%	6.7%	1.2%	1.3%	2.7%
14	3.1%	3.6%	2.9%	0.9%	1.2%	1.9%	1.9%	1.9%	5.3%	1%	1.4%	2.7%
15	2.8%	2.3%	1.8%	1.2%	1.1%	1.8%	1.5%	1.6%	3.9%	1.2%	1.8%	2.9%
16	2.8%	3.1%	2.9%	1.2%	1.2%	1.6%	1.3%	1.4%	3.9%	1.2%	1.8%	2.4%
17	3.4%	3.1%	1.8%	0.9%	1.5%	1.7%	1.5%	1.5%	2.9%	1.3%	1.3%	2%
18	3.8%	2.9%	2.3%	0.9%	1.7%	1.5%	1.9%	1.2%	2.9%	1.3%	1.1%	1.9%
19	3.1%	1.6%	1.9%	1.2%	1.4%	1.7%	1.7%	1.6%	2.4%	1.3%	1.4%	2.5%
20	2.8%	1.5%	1.9%	1.1%	1%	1.3%	1.6%	1.4%	2.1%	1.2%	1.3%	2.2%
21	2.5%	2.4%	1.8%	1.1%	1.1%	1.5%	1.6%	1.1%	2%	1%	1.2%	2.5%
22	3.2%	2.7%	1.3%	1.2%	1%	1.8%	1.8%	1.3%	2.4%	0.9%	1.3%	2.1%
23	3.2%	2.2%	1.1%	1.2%	0.8%	1.4%	2.4%	1.5%	2.3%	1%	1.3%	2.7%
24	1.8%	2.9%	1.1%	1.1%	1.4%	1.2%	1.4%	1.7%	1.7%	1.3%	1.1%	2%
25	3.7%	3.4%	1.2%	1.2%	1%	1.6%	1.4%	1.6%	2.1%	1.5%	0.8%	2.3%
26	3%	2.2%	1.2%	1.2%	0.8%	2%	1.7%	1.6%	2%	1.7%	0.8%	2.5%
27	3.1%	1.4%	1.4%	0.9%	0.9%	2.2%	2.1%	1.4%	1.8%	1.4%	0.8%	2.3%
28	2.7%	1.8%	1.3%	1.1%	0.8%	1.8%	2%	1.6%	1.4%	1.3%	0.9%	2.6%
29	3.3%	1.3%	1.1%	0.9%	0.9%	1.6%	1.7%	1.9%	1.4%	1.1%	1.1%	2.3%
30	2.8%		1.3%	1.3%	1.3%	1.4%	1.6%	2%	1.1%	1.2%	1.4%	2.9%
31	2.3%		1.7%		1.3%		1.7%	1.9%		1.5%		2.2%

Why is it a *good idea* to graph your data?

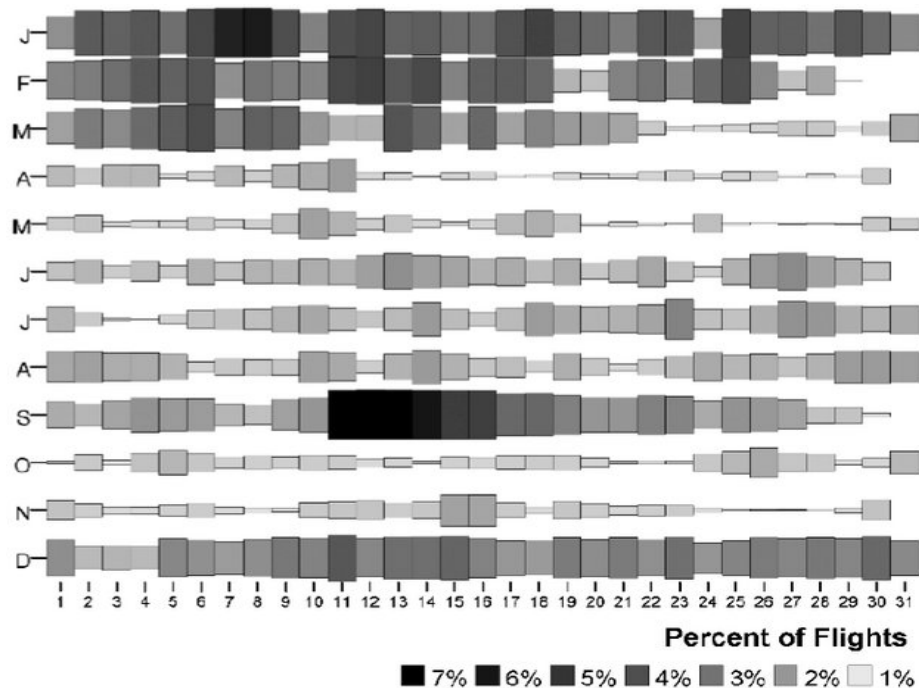


Figure 1. Percentage of flights canceled by day of year. The size is proportional to the absolute counts of flight canceled. Note the immediate outliers following September 11; one year of data where all flights were canceled is enough to make these dates look worse than they should.

- How many dimensions do you see?
 - This plot is a creative way to visualize the data, using both size and color as well as natural temporal ordering
- What are some takeaways?

Why is it a *good idea* to graph your data?

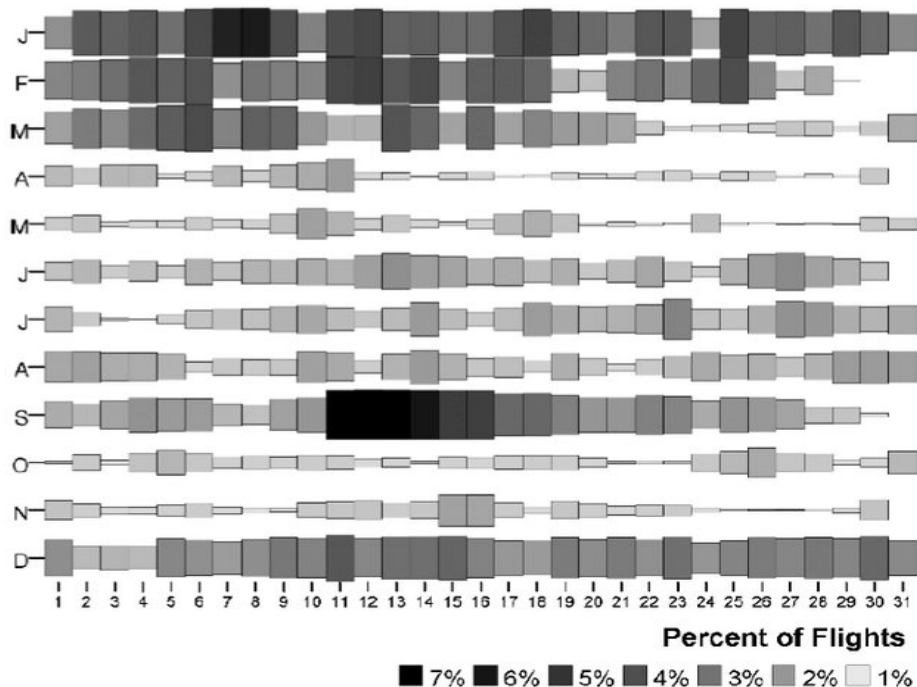
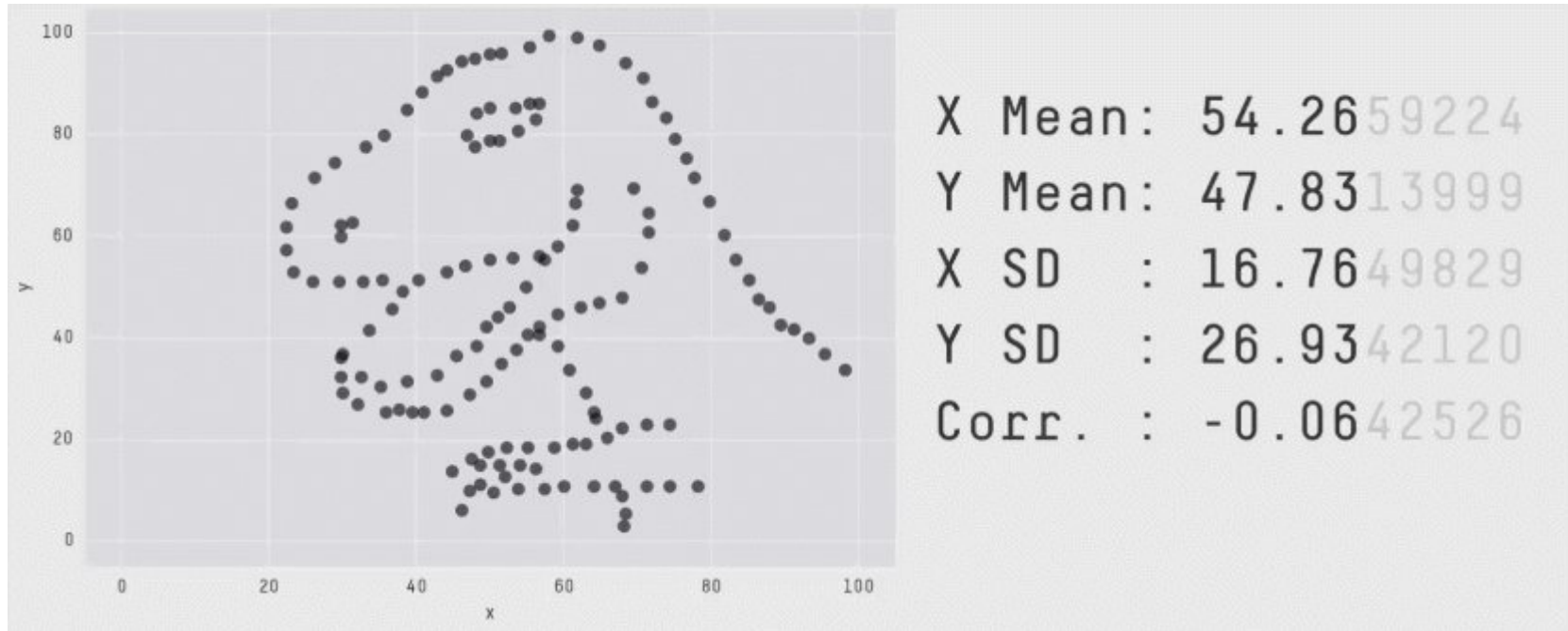


Figure 1. Percentage of flights canceled by day of year. The size is proportional to the absolute counts of flight canceled. Note the immediate outliers following September 11; one year of data where all flights were canceled is enough to make these dates look worse than they should.

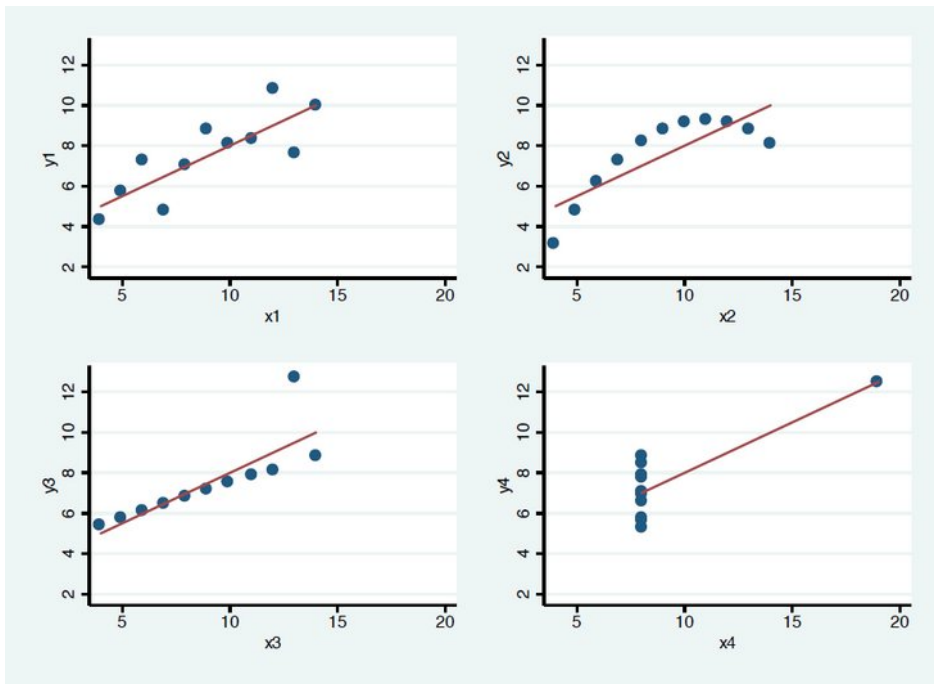
- How many dimensions do you see?
 - This plot is a creative way to visualize the data, using both size and color as well as natural temporal ordering
- What are some takeaways?
 - Winter cancellations as well as those of September 11, 2001 and the subsequent days stand out prominently
 - Good weather months recede from prominence
 - Advice: Fly in April, May, October, and November
 - Everyone gets home on Thanksgiving!

Why is it **critical** to graph your data? The Datasaurus...



Many data distributions have the same basic stats.

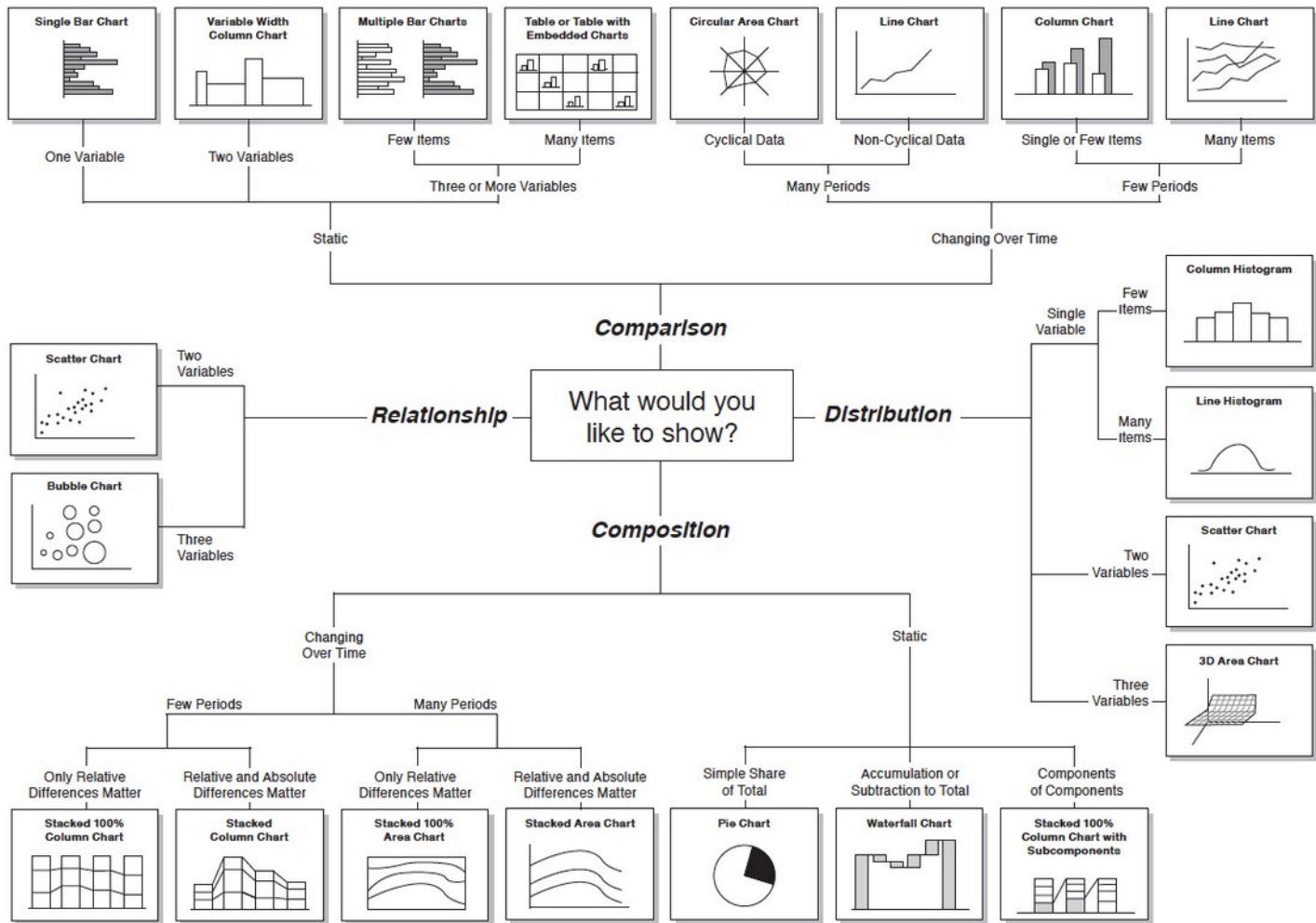
Why is it **critical** to graph your data?



- The first regression is an ideal regression.
- The second regression requires a nonlinear transformation.
- The third regression has a single large (vertical) outlier which has a large residual. This point is not very influential on the fitted line.
- The fourth regression has a large (horizontal) outlier (also called a leverage point). This point is extremely influential on the fitted line.

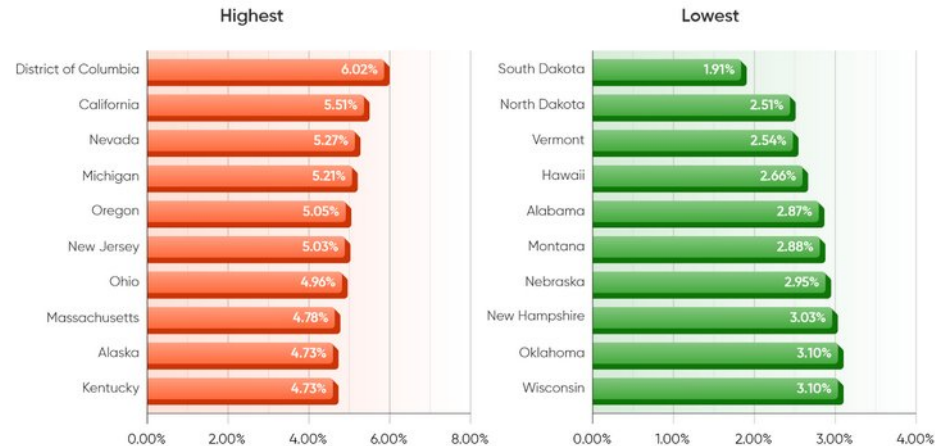
Chart Types

<https://extremepresentation.typepad.com/files/chart-chooser-2020.pdf>



Comparison: Static – Bar Chart

States with the highest and lowest unemployment rates



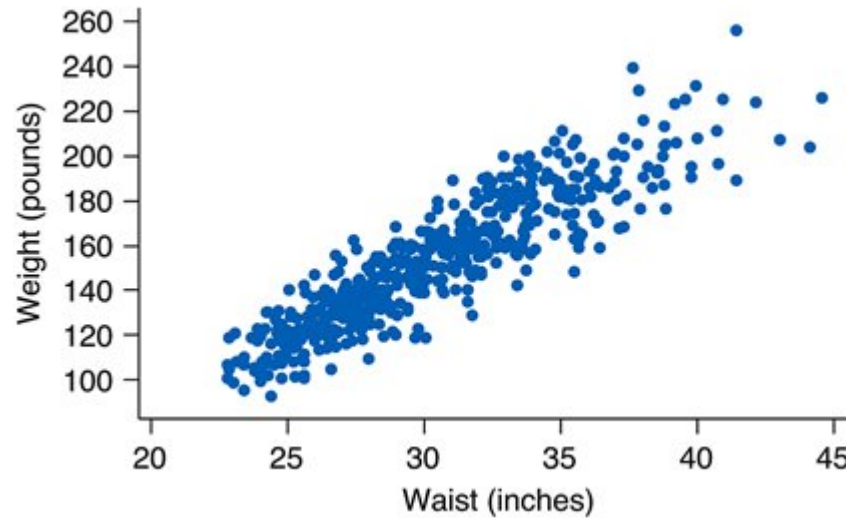
Data source: U.S. Bureau of Labour Statistics

Presented by:
 tradingpedia

Comparison: Trend over Time – Line Chart



Relationship: Scatterplot



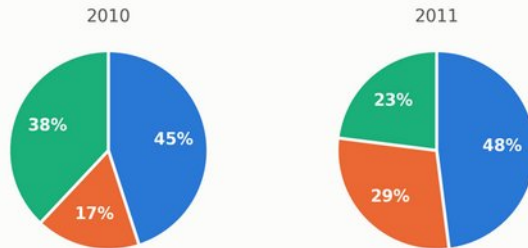
Composition: Pie Chart

Same numbers, three charts

Share of basket dollars by promotion level. The question: how did the mix change from 2010 to 2011?

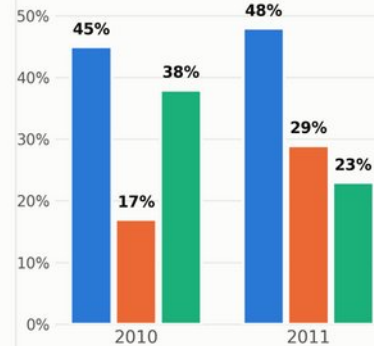
■ EDV (everyday value) ■ P1 (light promo) ■ P2 (deep promo)

Pie charts



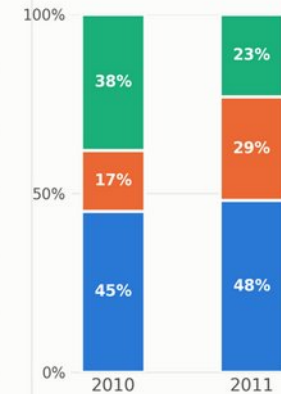
You are comparing angles across two circles.
Did P1 grow by 5 points or by 12? No way to tell.

Bar chart



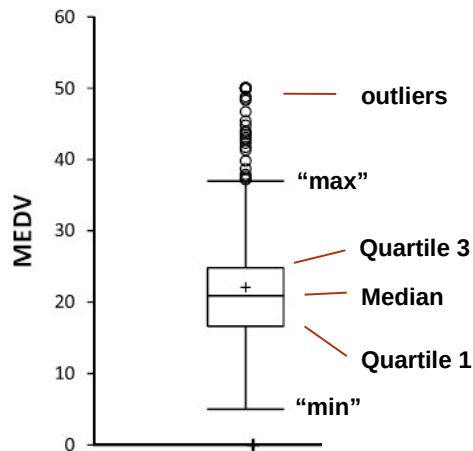
One shared baseline, so the change
reads straight off: P1 +12 points,
P2 -15, EDV +3.

Stacked bar

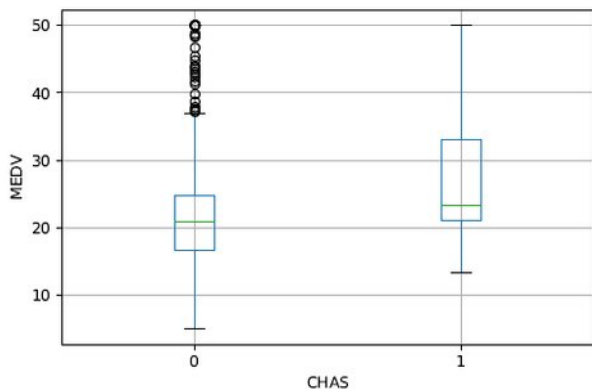


Parts of 100%, but only the
bottom segment sits on a
common baseline.

Distribution: Box Plot



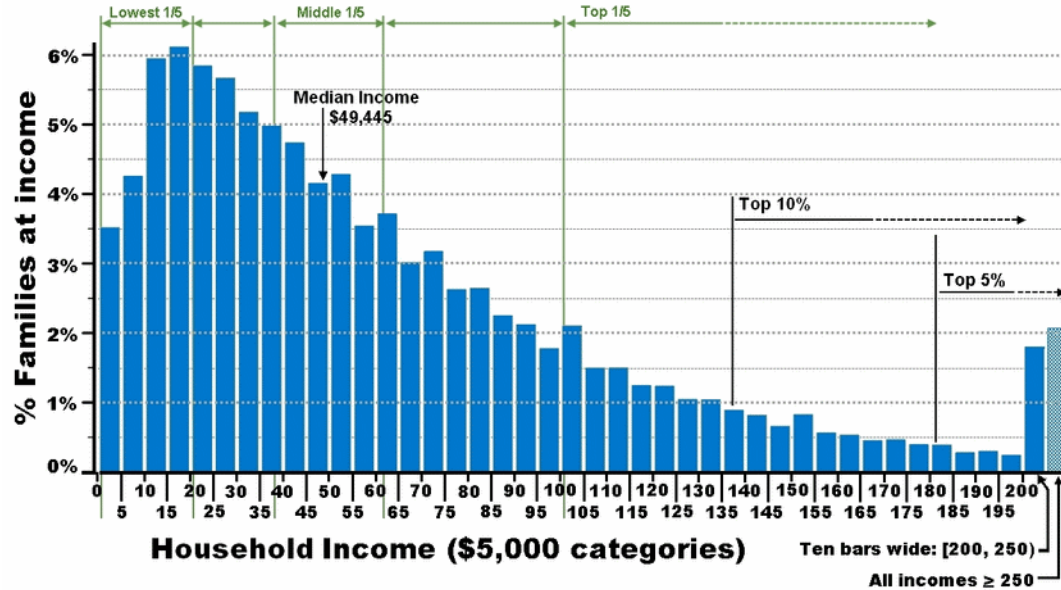
- Top outliers defined as those above $Q3 + 1.5(Q3 - Q1)$.
- “max” = maximum of non-outliers
- Analogous definitions for bottom outliers and for “min”
- Details may differ across software



Side-by-side boxplots are useful for comparing subgroups

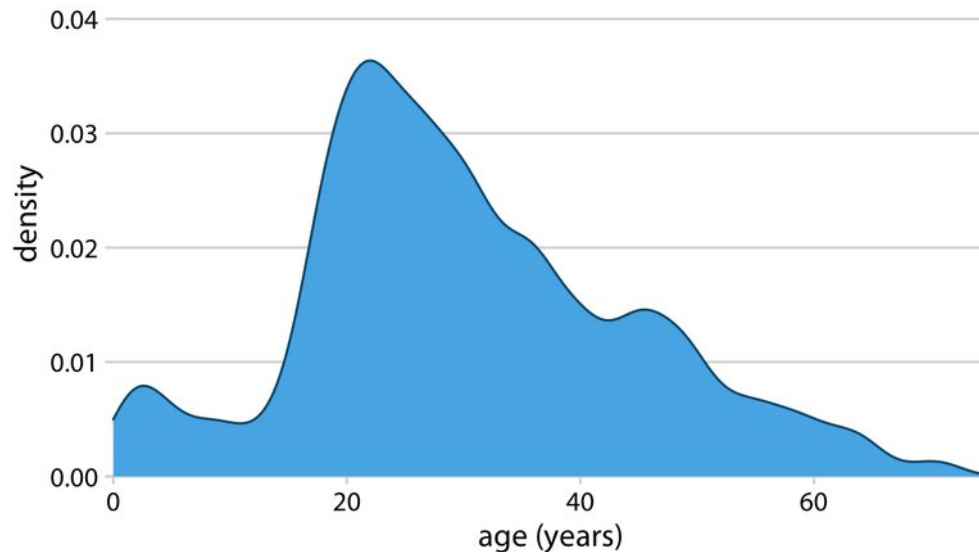
Houses in neighborhoods on Charles river (1) are more valuable than those not (0).

Distribution: Few Items – Histogram

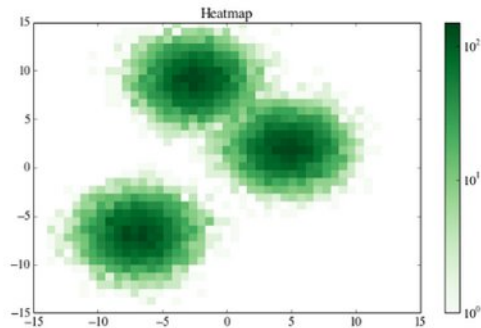
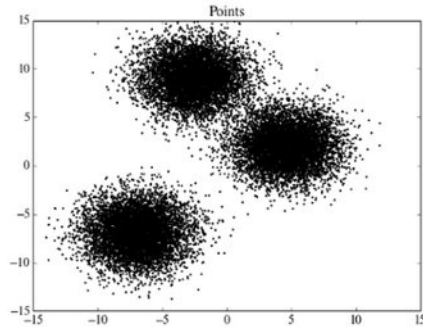


Data source: http://www.census.gov/hhes/www/cpstables/032011/hhinc/new06_000.htm

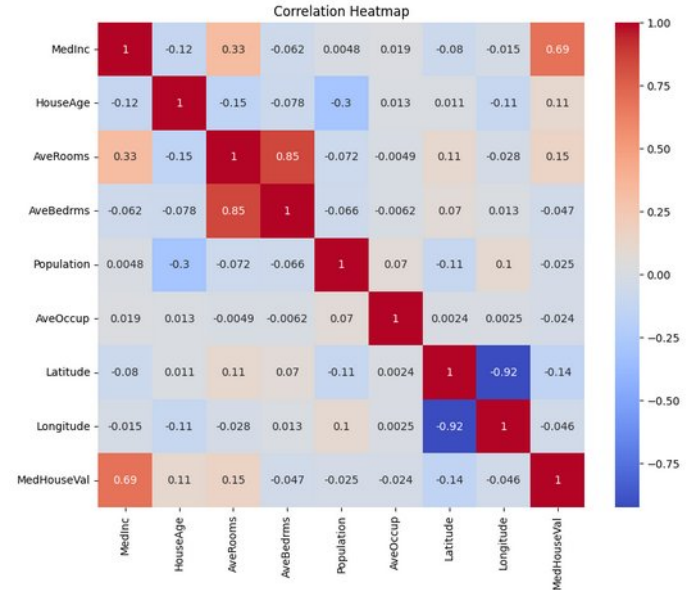
Distribution: Many Items – 1D Density Plot



Distribution: 2D Density Plot (Heat Map)

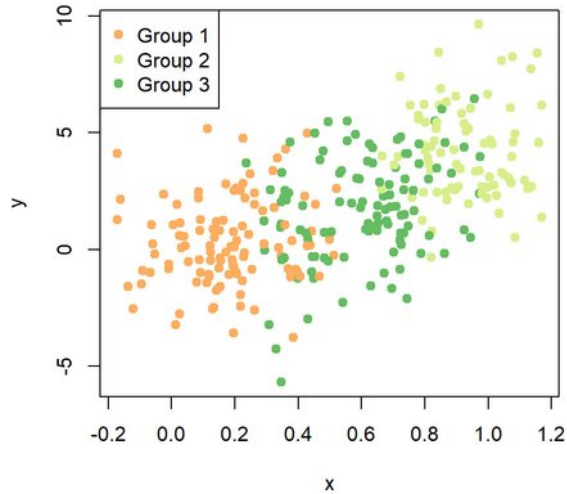


Heatmap to highlight correlations
Redder = stronger positive correlation
Bluer = stronger negative correlation

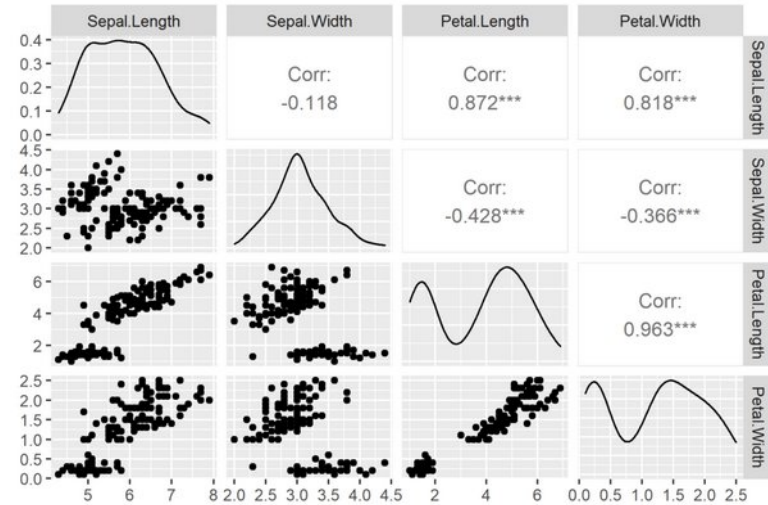


Multidimensional Visualization

Scatterplot with Color

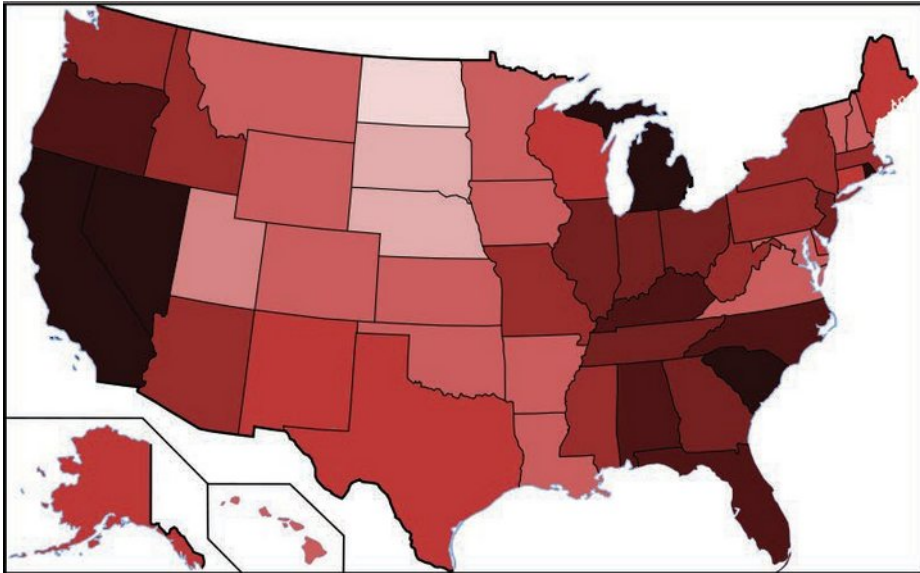


Matrix Scatterplot

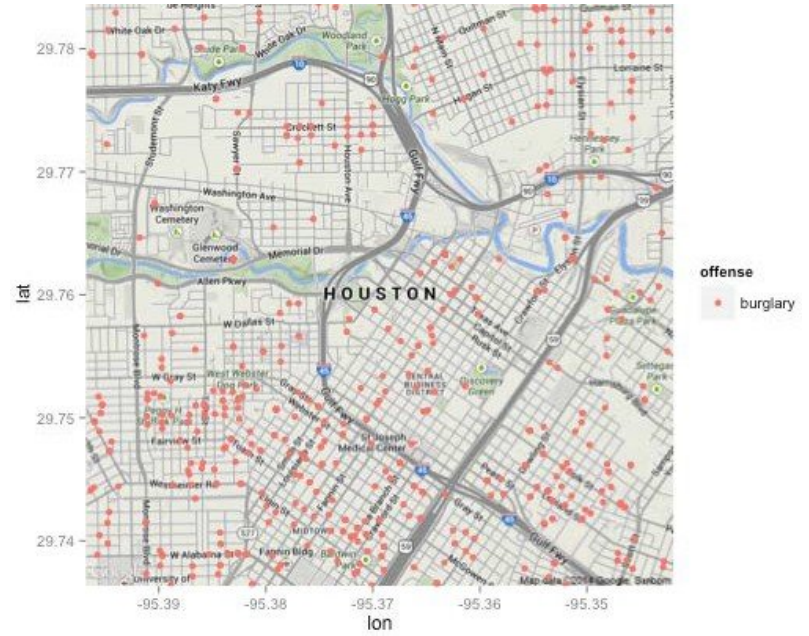


Geographical Data

Color a Map According to Data



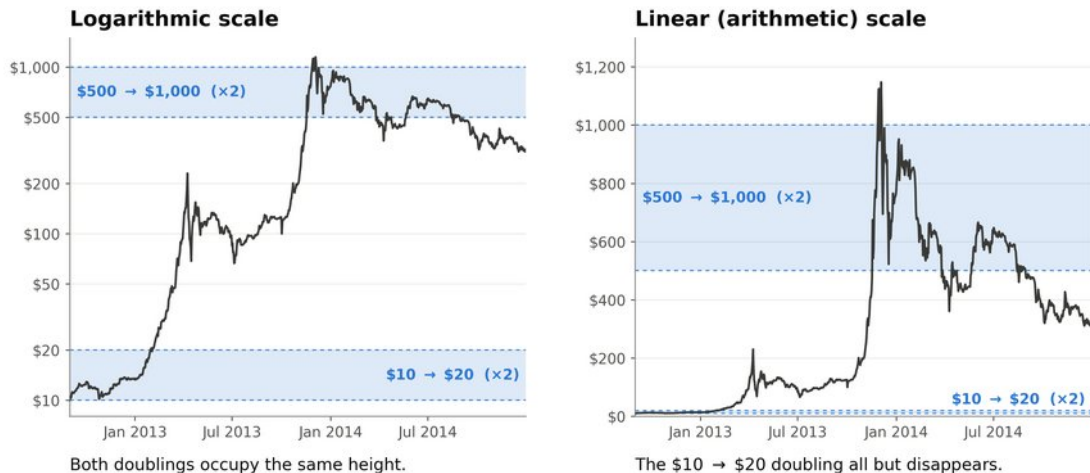
Add Geographical Data to a Map



Visual Manipulation: Rescaling, Aggregation, Zooming

Same data, two y-axes: on a log scale, equal percentage moves have equal height

Bitcoin, daily close in USD, Sep 2012 – Dec 2014. Shaded bands mark two +100% moves.



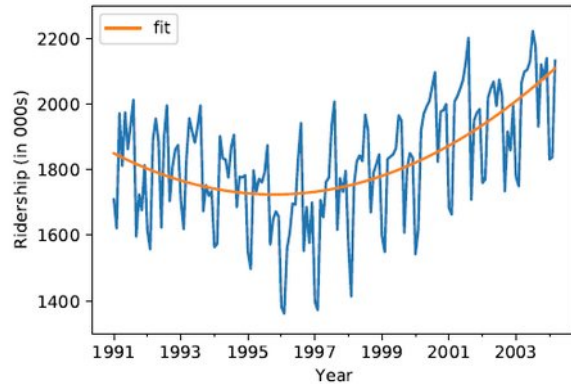
Both doublings occupy the same height.

The \$10 → \$20 doubling all but disappears.

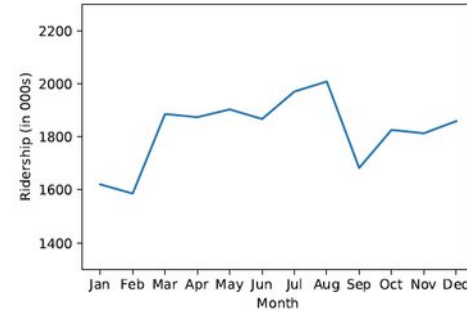
Data: daily BTC/USD close (Habrador/Bitcoin-price-visualization, GitHub).

Visual Manipulation: Rescaling, Aggregation, Zooming

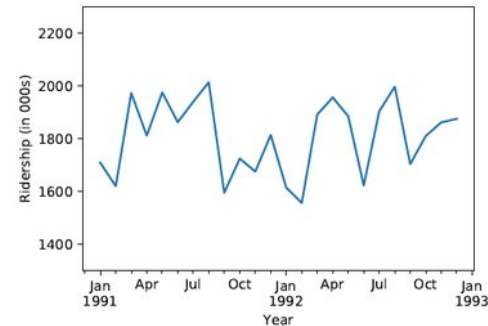
Amtrak Ridership
Monthly Data – Curve Added



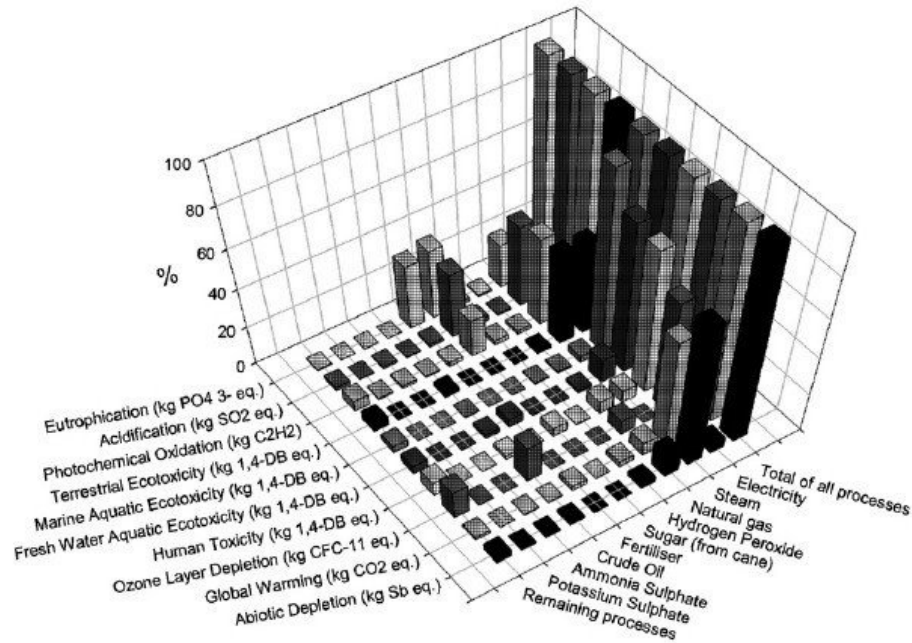
Monthly Averages



Zoom In



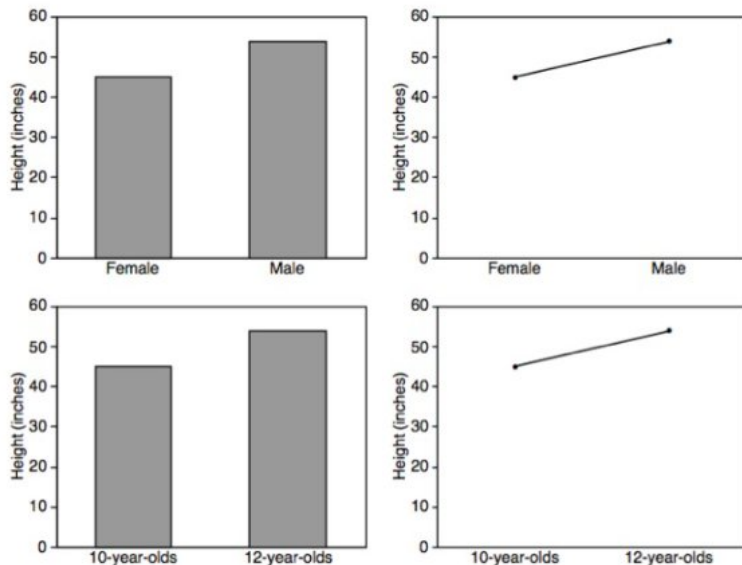
Don't Go 3D



Don't Use Trend Lines for Categorical Data

Bars vs. Lines

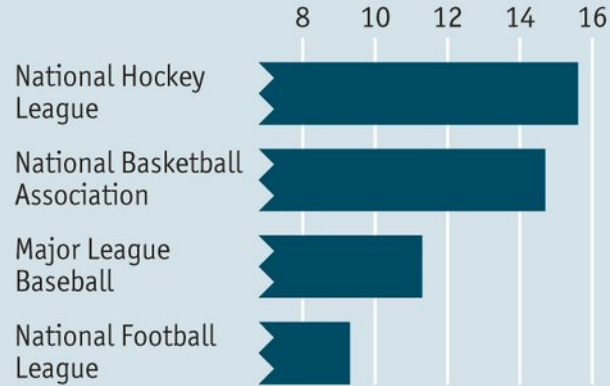
Lines imply connections - do not use for categorical data



Don't Break the Y-axis Scale

No room to run

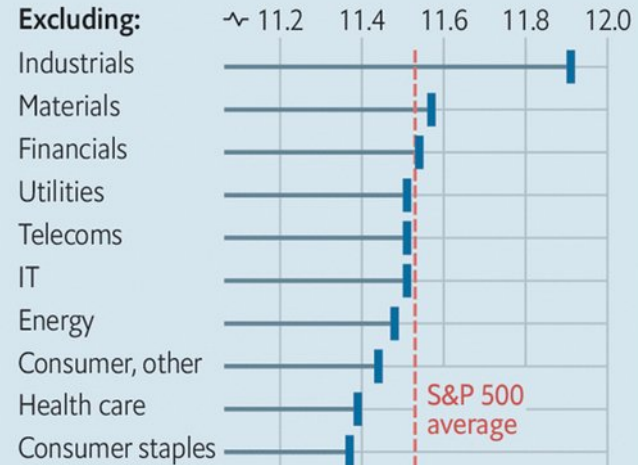
Annual average salary growth since 1990, %



Source: John Vrooman

A narrow exclusion zone

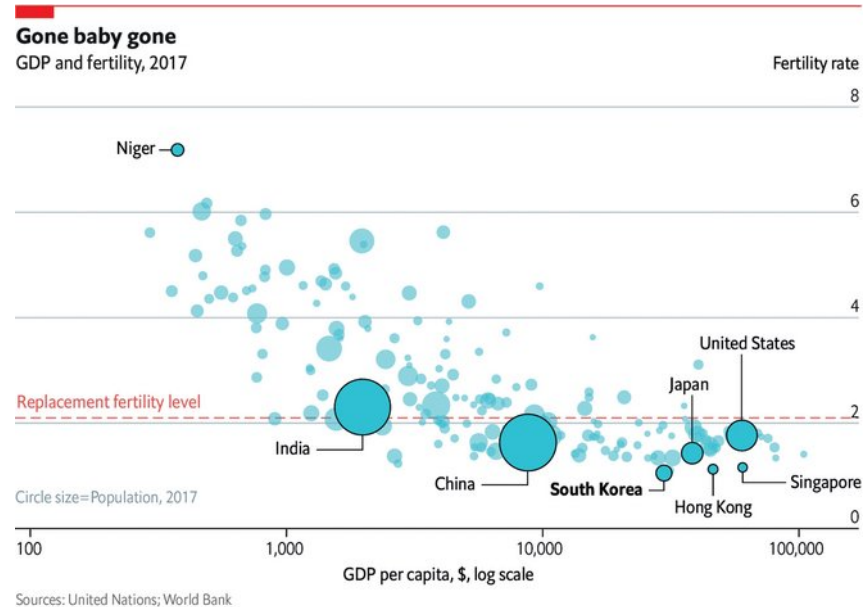
S&P 500, average annual total returns, %
1925-2017*



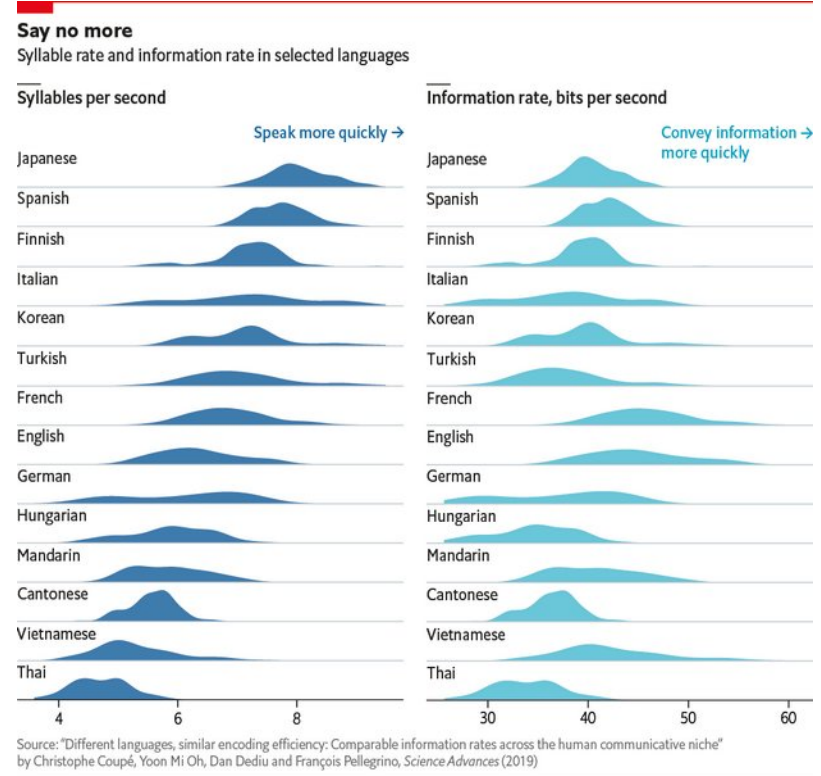
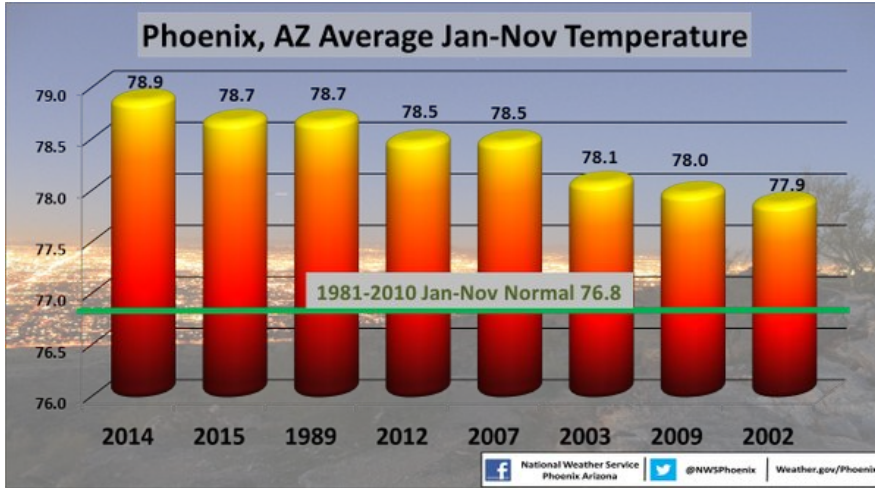
Sources: S&P; GMO

*S&P 90 prior to March 1957

Don't Crowd Your Charts

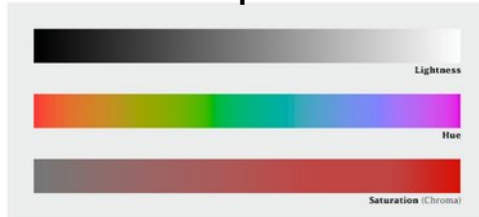


Don't Crowd Your Charts



Use Color Scales Wisely

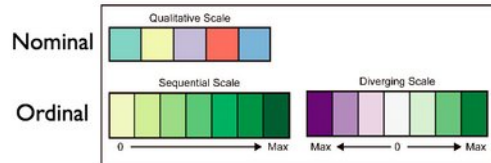
Color Space



Rainbow colormap is perceptually nonlinear



Brewer Scales

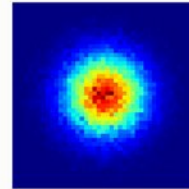


Pick the color scale from the data

The scale is part of the encoding, not decoration. Two kinds of numbers, two rules — and one scale that

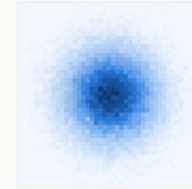
SEQUENTIAL · a count, 0 and up, one direction → one hue, light to dark

✗ Rainbow (jet)



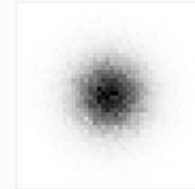
Lightness climbs, then falls. The yellow ring reads as a boundary — there is no boundary in the data.

✓ One hue, light → dark



Darker = more, and nothing else changes. The eye ranks the values without being taught the key.

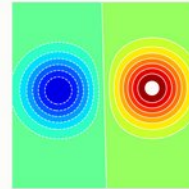
✓ Grey, light → dark



The same ramp on one channel. Survives photocopying and every form of color blindness.

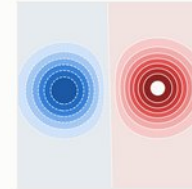
DIVERGING · a signed value with a real zero → two hues, neutral in the middle

✗ Rainbow (jet)



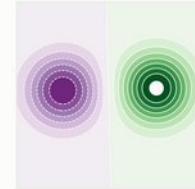
Zero has no anchor: it lands somewhere inside the green. Sign is guesswork without reading the key.

✓ Blue - grey - red



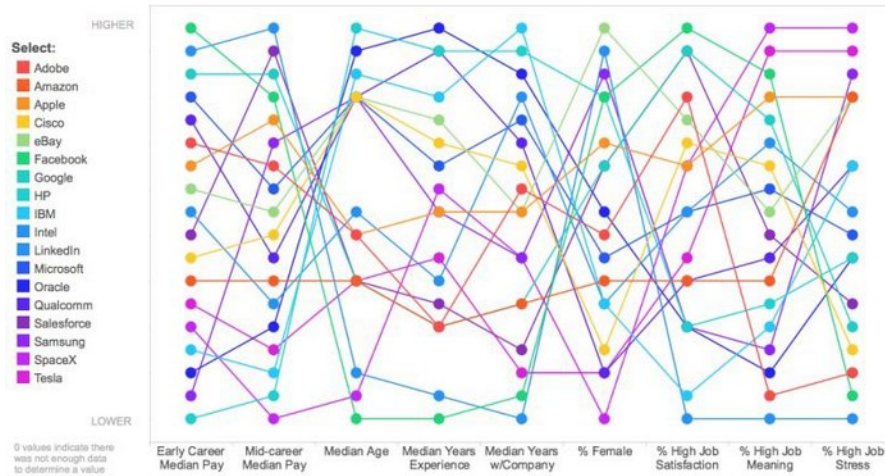
Neutral at zero, one hue per sign, equal arms — so -0.3 and $+0.3$ look equally far from nothing.

✓ Purple - white - green



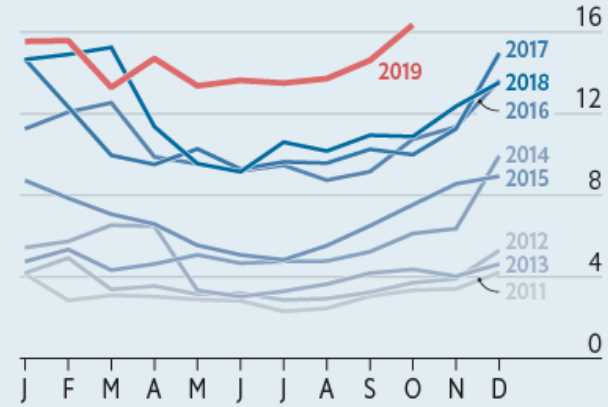
Same structure, different hues. Use it when "red = bad" would say something about the data that isn't true.

Use Fewer than Seven Colors



Not just a winter crisis

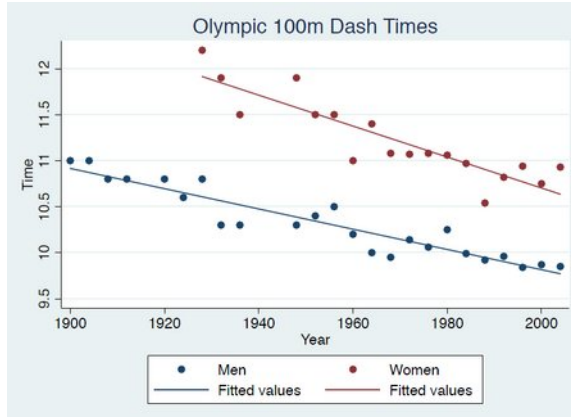
England, patients waiting* more than four hours in hospital emergency departments, %



Source: NHS England

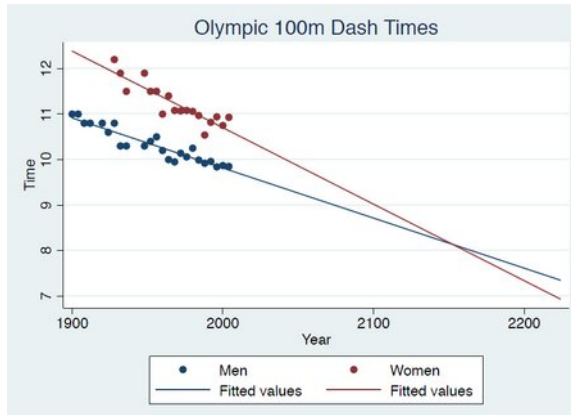
*From arrival to admission, transfer or discharge

Don't Extrapolate Well Beyond Data Limits

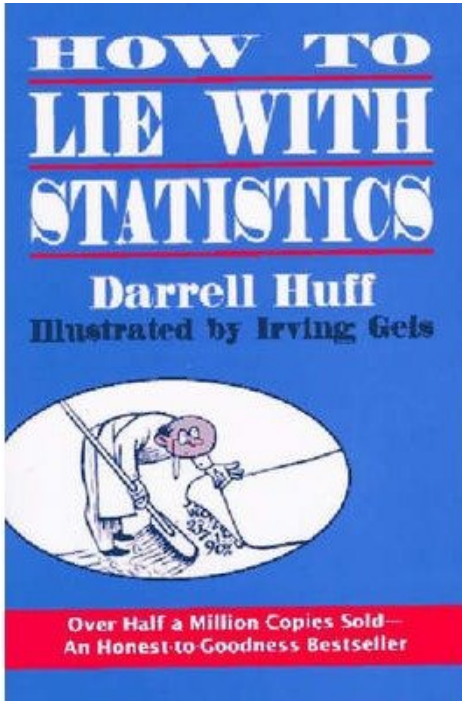


Nature, 2004 — "Momentous sprint at the 2156 Olympics?"

- Olympic 100m times, men and women, fit by linear regression. $R^2 = 0.88$ and 0.79 . Excellent fit.
- The lines converge and cross at the **2156 Olympics**: women 8.079 s vs. men 8.098 s (95% CI: 2064–2788).
- Keep following the line: **7.0 s by 2224** (~32 mph) → **sound barrier by 2600** → eventually faster than light, finishing before the gun.
- The model knows only "times went down." It knows nothing about muscle, physics, or the fact that real gains flatten.
- **R^2 measures fit to the data you have. It says nothing about the data you don't.**



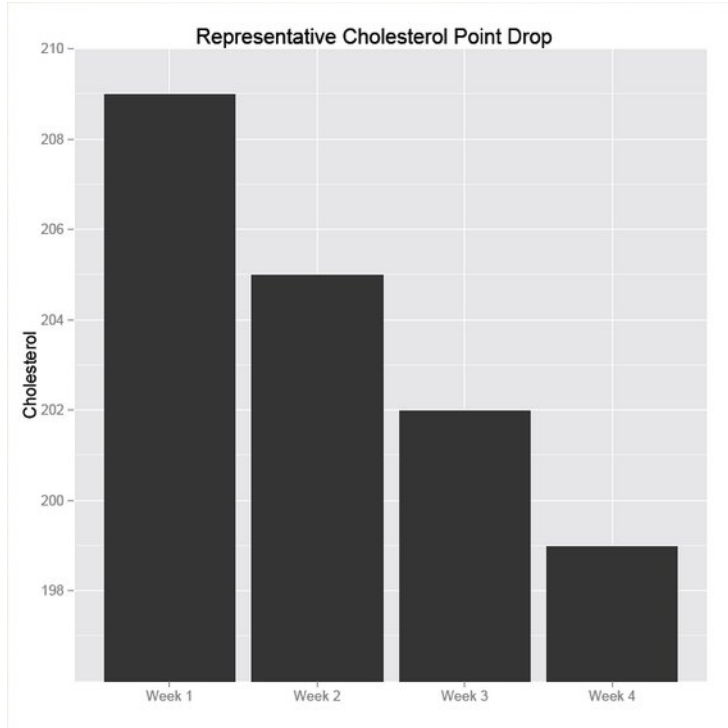
Lying with Statistics



Dark arts of statistical deception via graphics

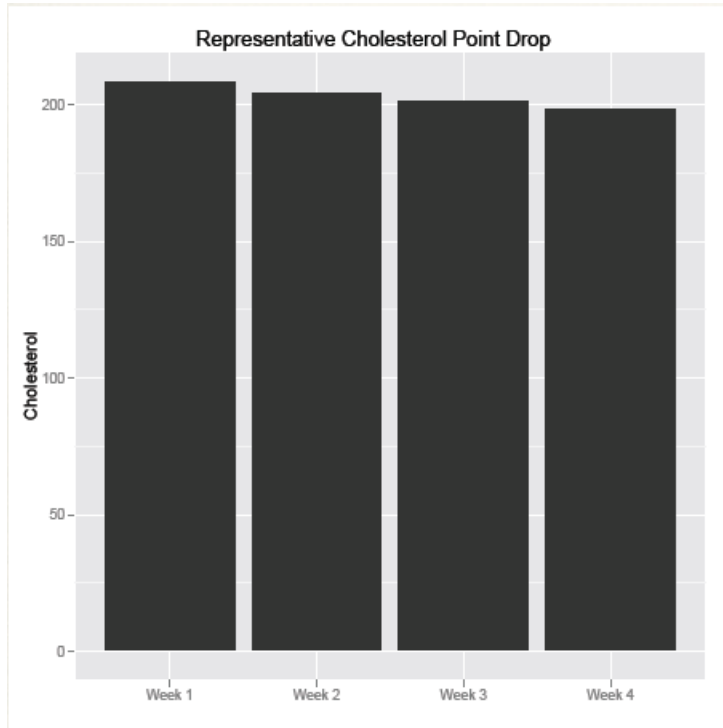
- Don't use these techniques on any of your projects!
- Do beware of them when reading reports, the news, and in your own writing.

How to Lie with Statistical Graphics - 1



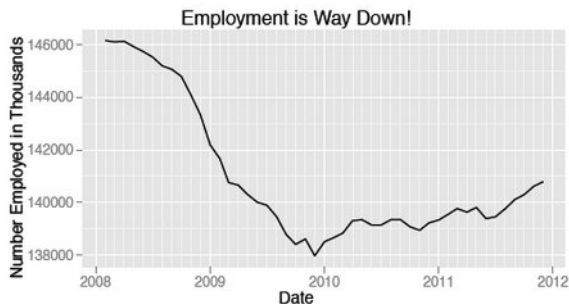
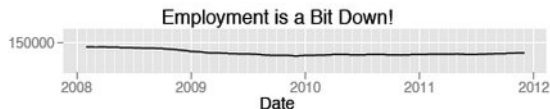
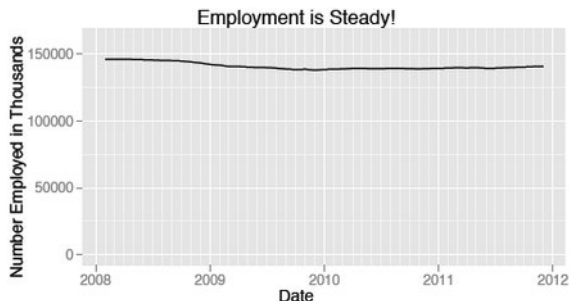
- Quaker Oats launched a marketing campaign to convince the public that their product reduces cholesterol.
- They used this bar plot to emphasize the miraculous effects of their bland oats.
- Eat Quaker Oats and your cholesterol levels will drop dramatically!

How to Lie with Statistical Graphics - 1



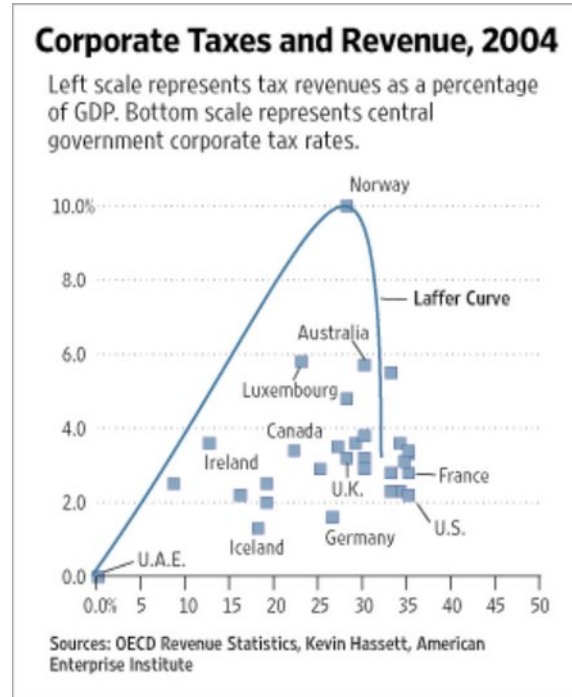
- Bar plots convey meaning through area (which in this case is proportional to length since all bars are of the same width).
- By not starting at zero, Quaker made it seem like their oatmeal was having a dramatic effect when it was not.
- After reams of complaints, Quaker withdrew the plot from the campaign materials.

How to Lie with Statistical Graphics - 2



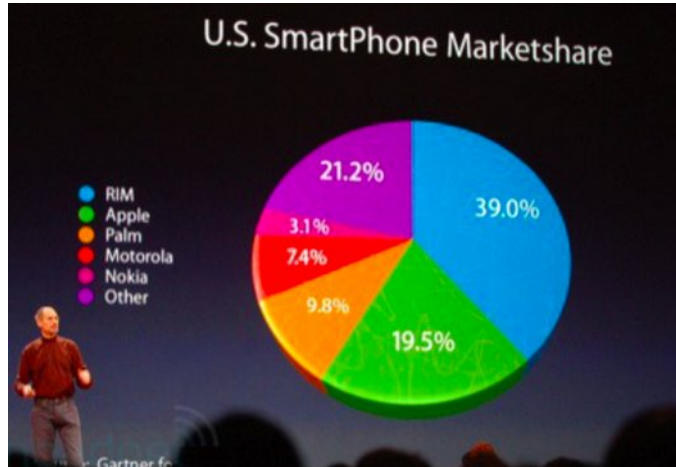
- Axis scales suggest comparisons.
- Even with no malintent on the part of the graph's author, these three plots of exactly the same data communicate very different messages.
- Carefully consider what message(s) you are trying to convey.

How to Lie with Statistical Graphics - 3

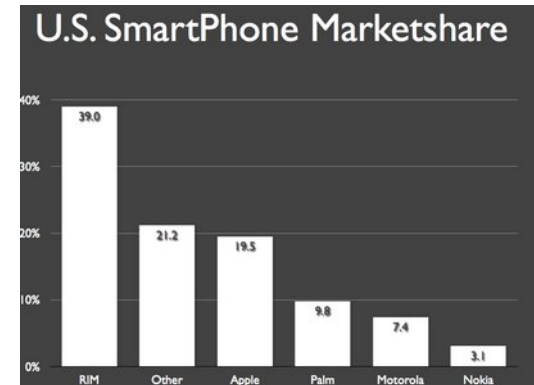
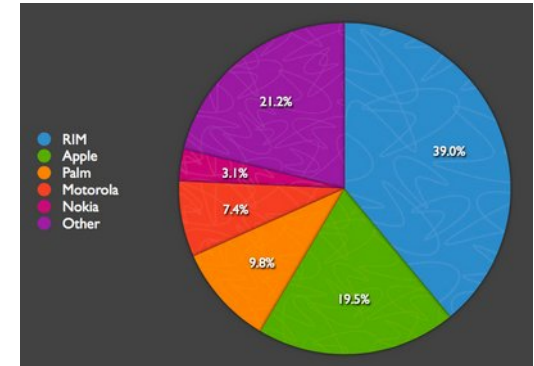


- The Wall Street Journal editorial board published this graph with a Laffer Curve superimposed.
- It clearly demonstrates that if you increase tax rates beyond a certain point, then tax revenues will drop.
- The problem is that there appears to be no such pattern in the data.

How to Lie with Statistical Graphics - 4



VS.



The representation of numbers, as physically measured on the surface of the graphic itself, should be directly proportional to the quantities represented.

[Tufte, 1991]

This results in the following formula:

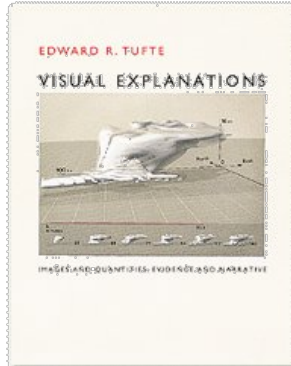
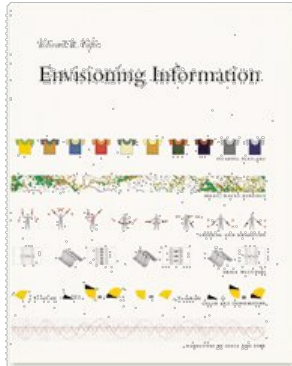
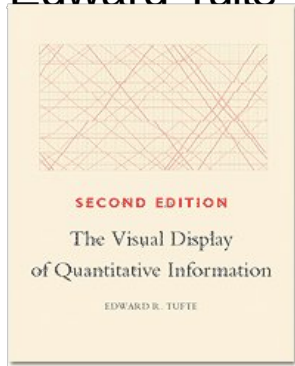
$$\text{Lie Factor} = \frac{\text{size of effect shown in graphic}}{\text{size of effect in data}}$$

Data Visualization Final Tips

- Every graph should **tell a story**
 - What are you trying to say with the graph?
 - What comparisons are you trying to make?
- Stories require characters
 - Give every graph a **title**.
 - Make sure all **axes are titled, labeled, and legible**.
 - Provide **units**.
 - Make **palettes** and symbols distinct.
 - Provide **legends** for the data series or, even better, directly label them.
- A statistical graph does not need to stand alone
 - It is often enhanced by text.
 - Provide a **caption to explain the graph**--not merely what the points and lines indicate but also what messages the graph sends.
- Good plots maximize the data-ink ratio and minimize chartjunk
 - Highlight the most important comparisons and tell a story
 - Often using text as well as graphics.
- Pie charts, Excel graphics, and tables generally should not be used.
- Don't lie with or be fooled by statistical graphics!

Further Reading

Edward Tufte



Stephen Few

